Example 4.5 A person wants to buy a car. There are two brands of car available in the market and each brand has 3 variant models and each model comes in five different colours as in Figure 4.2 In how many ways she can choose a car to buy?

Solution:
A car can be bought by choosing a brand, then a variant model, and then a colour. A brand can be chosen in 2 ways; a model can be chosen in 3 ways and a colour can be chosen in 5 ways. By the rule of product the person can buy a car in $2 \times 3 \times 5 = 30$ different ways.

Example 4.6 A Woman wants to select one silk saree and one sungudi saree from a textile shop located at Kancheepuram. In that shop, there are 20 different varieties of silk sarees and 8 different varieties of sungudi sarees. In how many ways she can select her sarees?

Solution:
The work is done when she selects one silk saree and one sungudi saree. The Woman can select a silk saree in 20 ways and sungudi saree in 8 ways. By the rule of product, the total number of ways of selecting these 2 sarees is $20 \times 8 = 160$.

Example 4.7 In a village, out of the total number of people, 80 percentage of the people own Coconut groves and 65 percent of the people own Paddy fields. What is the minimum percentage of people own both?

Solution:
Let $n(C)$ denote the percentage of people who own the Coconut groves and $n(P)$ denote the percentage of people who own Paddy fields. We are given $n(C) = 80$ and $n(P) = 65$. By the rule of inclusion - exclusion $n(C \cap P) = n(C) + n(P) - n(C \cup P)$. The maximum value of $n(C \cup P)$ is 100. Therefore, the minimum value of $n(C \cap P)$ is $80 + 65 - 100 = 45$. That is, the minimum percentage of the people who own both is 45.



In the next problem, we use the notion of a 'string'. A string is formed by writing given letters one by one in a sequence. For instance, strings of length three formed out of the letters a,b,c & d are aaa, abb, bda, dca, cdd $\cdots$.

Example 4.8

(i) Find the number of strings of length 4, which can be formed using the letters of the word BIRD, without repetition of the letters.
(ii) How many strings of length 5 can be formed out of the letters of the word PRIME taking all the letters at a time without repetition.

Solution:

(i) There are as many strings as filling the 4 vacant places by the 4 letters, keeping in mind that repetition is not allowed. The first place can be filled in 4 different ways by any one of the letters B,I,R,D. Following which, the second place can be filled in by any one of the remaining 3 letters in 3 different ways, following which the third place can be filled in 2 different ways, following which fourth place can be filled in 1 way.